

Lab 10: Set Up an App Dev Environment - Challenge Lab

Challenge Lab | Google Cloud Associate Cloud Engineer | GSP315

 **This is a Challenge Lab — no step-by-step instructions provided.**
Skills from Labs 01-09 were applied independently to complete all tasks.

Overview

Applied all skills from the Cloud Storage and IAM Foundations badge independently. Helped the fictional Memories team set up their app dev environment by creating a storage bucket, Pub/Sub topic, Cloud Run Function for image thumbnails, and removing a previous engineer's access.

Task 1 - Create a Storage Bucket

Created a Cloud Storage bucket named per the lab requirements in the specified region to store photographs for the Memories project.

- Bucket name: [BUCKET NAME from lab]
- Region: [REGION from lab]
- Used skills from Labs 01 and 02

```
gcloud storage buckets create gs://[BUCKET-NAME] --region=[REGION]
```

The screenshot shows the Google Cloud Storage console interface. At the top, there's a header with the Google Cloud logo, a project ID 'qwiklabs-gcp-00-db99f13477de', and navigation icons. Below the header, the 'Cloud Storage' section is active. A 'Buckets' tab is selected, with 'Create' and 'Refresh' buttons. A 'Filter' button is also present. The main area displays a table of buckets:

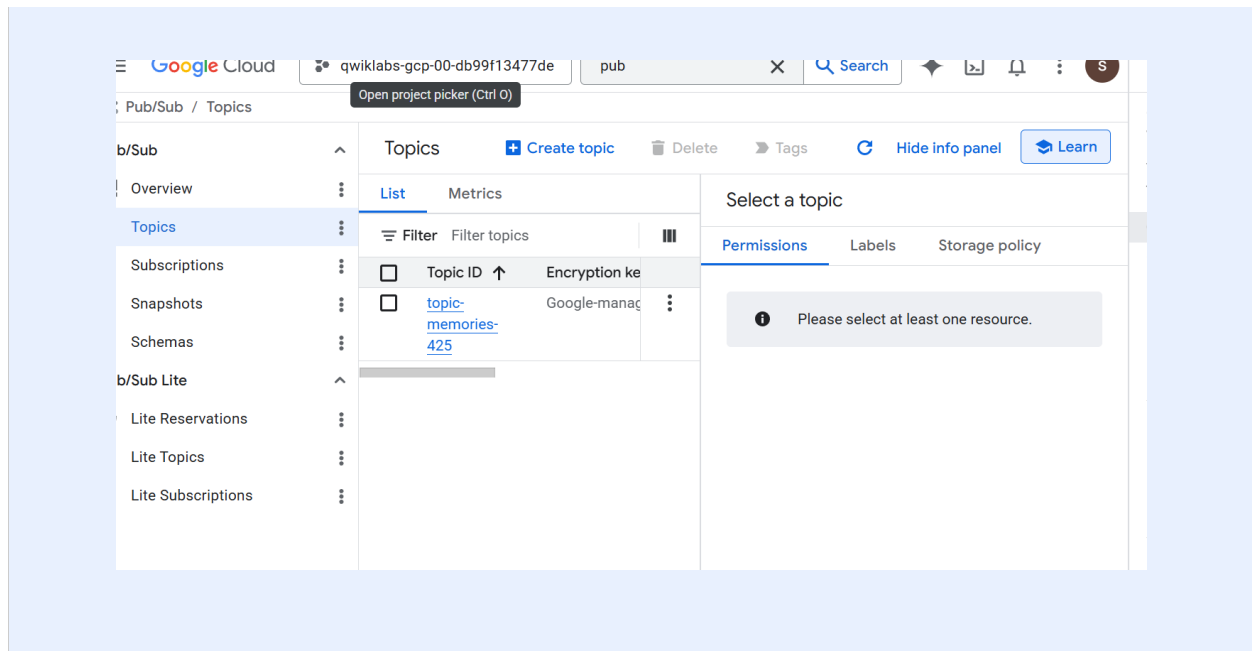
☐	Name ↑	Created	Location type	Location
☐	gcf-v2-sources-890169214288-us-ce...	May 25, 2026, 8:10:54 PM	Region	us-central1
☐	gcf-v2-uploads-890169214288.us-ce...	May 25, 2026, 8:10:12 PM	Region	us-central1
☐	qwiklabs-gcp-00-db99f13477de-bucket	May 25, 2026, 8:04:53 PM	Region	us-central1

Task 2 - Create a Pub/Sub Topic

Created a Pub/Sub topic that the Cloud Run Function will use to send messages when a thumbnail is successfully created.

- Topic name: [TOPIC NAME from lab]
- Used skills from Labs 07 and 08

```
gcloud pubsub topics create [TOPIC-NAME]
```



Task 3 - Create the Thumbnail Cloud Run Function

Created a 2nd generation Cloud Run Function triggered by Cloud Storage uploads. The function automatically creates a 64x64 thumbnail of any PNG or JPG image uploaded to the bucket and publishes a message to the Pub/Sub topic.

- Function name: [FUNCTION NAME from lab]
- Runtime: Node.js 22
- Generation: 2nd generation
- Trigger: Cloud Storage - object created
- Entry point: [FUNCTION NAME from lab]
- Bucket: [BUCKET NAME from lab]

index.js code:

```
const functions = require('@google-cloud/functions-framework');
const { Storage } = require('@google-cloud/storage');
const { PubSub } = require('@google-cloud/pubsub');
const sharp = require('sharp');
```

```
functions.cloudEvent('[FUNCTION-NAME]', async cloudEvent => {
  const event = cloudEvent.data;
  const fileName = event.name;
  const bucketName = event.bucket;
  const size = '64x64';
  const bucket = new Storage().bucket(bucketName);
  const topicName = '[TOPIC-NAME]';
  const pubsub = new PubSub();
```

```
  if (fileName.search('64x64_thumbnail') === -1) {
    const filename_split = fileName.split('.');
    const filename_ext = filename_split[filename_split.length - 1].toLowerCase();
```

```

    const filename_without_ext = fileName.substring(0, fileName.length -
filename_ext.length - 1);

    if (filename_ext === 'png' || filename_ext === 'jpg' || filename_ext ===
'jpeg') {
        const gcsObject = bucket.file(fileName);
        const newFilename =
`${filename_without_ext}_64x64_thumbnail.${filename_ext}`;
        const gcsNewObject = bucket.file(newFilename);

        const [buffer] = await gcsObject.download();
        const resizedBuffer = await sharp(buffer).resize(64, 64).toBuffer();
        await gcsNewObject.save(resizedBuffer);
        await pubsub.topic(topicName).publishMessage({ data: Buffer.from(newFilename)
});
    }
}
});

```

package.json code:

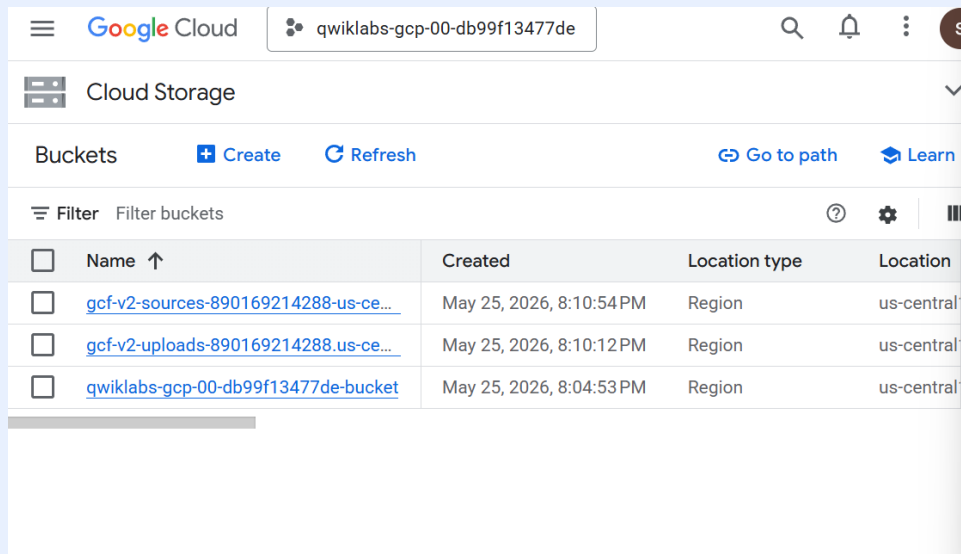
```

{ "name": "thumbnails", "version": "1.0.0",
  "dependencies": {
    "@google-cloud/functions-framework": "^3.0.0",
    "@google-cloud/pubsub": "^2.0.0",
    "@google-cloud/storage": "^6.11.0",
    "sharp": "^0.32.1"
  }
}

```

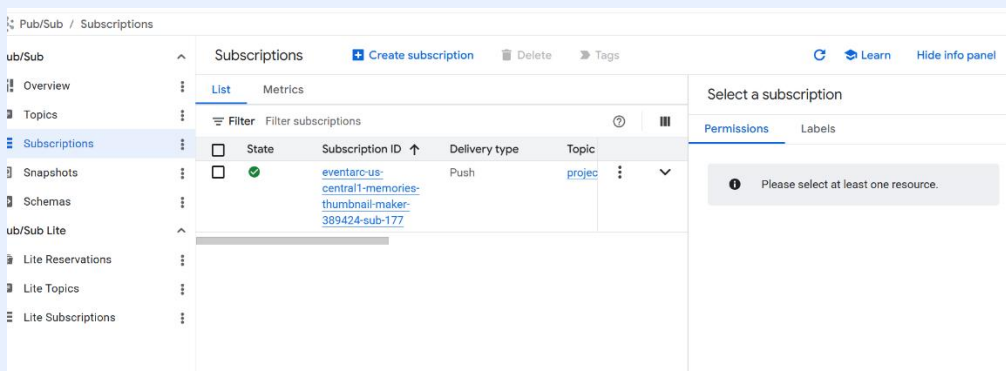
The screenshot shows the Google Cloud Platform console interface. At the top, there's a search bar and navigation links. The main section is titled 'Services' and includes a description: 'A service exposes a unique endpoint and automatically scales the underlying infrastructure to handle incoming requests. Deploy a container image, source code or a function to create a service.' Below this, there's a table of services. The table has columns for Name, Deployment type, Req/sec, Region, Cost, and Authentication. One service is listed: 'memories-thumbnail-maker', which is a Function deployment in the us-central1 region, with 0 requests per second and public access.

Name	Deployment type	Req/sec	Region	Cost	Authentication
memories-thumbnail-maker	Function	0	us-central1	—	Public access



The screenshot shows the Google Cloud Storage Buckets page. At the top, there's a search bar with the project ID 'qwiklabs-gcp-00-db99f13477de'. Below the header, there are buttons for 'Create' and 'Refresh', and links for 'Go to path' and 'Learn'. A filter bar is present above a table of buckets. The table has columns for Name, Created, Location type, and Location. Three buckets are listed, all in the 'us-central' location.

Name	Created	Location type	Location
gcf-v2-sources-890169214288-us-ce...	May 25, 2026, 8:10:54 PM	Region	us-central
gcf-v2-uploads-890169214288.us-ce...	May 25, 2026, 8:10:12 PM	Region	us-central
qwiklabs-gcp-00-db99f13477de-bucket	May 25, 2026, 8:04:53 PM	Region	us-central



The screenshot shows the Google Cloud Pub/Sub Subscriptions page. On the left is a sidebar with navigation options like Overview, Topics, Subscriptions, and Lite. The main area shows a table of subscriptions. One subscription is listed with a 'Push' delivery type and a topic named 'projec'. On the right, there's a panel for 'Select a subscription' with tabs for 'Permissions' and 'Labels'. A message at the bottom of this panel says 'Please select at least one resource.'

State	Subscription ID	Delivery type	Topic
☒	eventarc-us-central1-memories-thumbnail-maker-389424-sub-177	Push	projec

Task 4 - Remove Previous Cloud Engineer Access

Removed the Viewer role from Username 2 (the previous cloud engineer) using IAM. Applied skills from Lab 03.

- Navigation: IAM and Admin → IAM
- Found Username 2 with Viewer role
- Clicked pencil icon → trash icon to remove role
- Clicked Save to confirm

```
student_01_b0eb51674034@cloudshell:~/memories-thumbnail-maker (qwiklabs-gcp-00-db99f13477de) $ gcloud projects remove-iam-policy-binding qwiklabs-gcp-00-db99f13477de \
--member="user:student-02-e77c4b3a811f@qwiklabs.net" \
--role="roles/viewer" \
Updated IAM policy for project [qwiklabs-gcp-00-db99f13477de].
bindings:
members:
- serviceAccount:service-890169214288@gcp-sa-artifactregistry.iam.gserviceaccount.com
role: roles/artifactregistry.serviceAgent
members:
- serviceAccount:qwiklabs-gcp-00-db99f13477de@qwiklabs-gcp-00-db99f13477de.iam.gserviceaccount.com
role: roles/bigquery.admin
members:
- serviceAccount:890169214288@cloudbuild.gserviceaccount.com
role: roles/cloudbuild.builds.builder
members:
- serviceAccount:service-890169214288@gcp-sa-cloudbuild.iam.gserviceaccount.com
role: roles/cloudbuild.serviceAgent
members:
- serviceAccount:service-890169214288@gcf-admin-robot.iam.gserviceaccount.com
role: roles/cloudfunctions.serviceAgent
members:
- serviceAccount:890169214288@cloudservices.gserviceaccount.com
role: roles/compute.instanceGroupManagerServiceAgent
members:
- serviceAccount:service-890169214288@compute-system.iam.gserviceaccount.com
role: roles/compute.serviceAgent
members:
- serviceAccount:service-890169214288@container-analysis.iam.gserviceaccount.com
role: roles/containeranalysis.ServiceAgent
members:
- serviceAccount:service-890169214288@gcp-sa-containerscanning.iam.gserviceaccount.com
role: roles/containerscanning.ServiceAgent
```

You need additional access

You need additional access to the project:

• qwiklabs-gcp-00-db99f13477de

This could be because you have insufficient permissions to access the resource, or because a deny or principal access boundary policy is blocking your access to the resource. [Learn more](#)

Missing or blocked permissions ⓘ [Copy Troubleshooter URL](#)
resourcemanager.projects.get (Missing)

[Request access](#)

Applicable roles ⓘ

The following roles contain all of the missing required permissions for accessing this resource. Ordering is based on the type of actions permitted by the role, service relevance, and the number of permissions.

Project Mover

Role name	roles/resourcemanager.projectMover
Title	Project Mover
Description	Access to update and move a project
Permissions	View included permissions

What I Learned

- Challenge labs require combining multiple skills learned across all previous labs
- Cloud Run Functions can be triggered by Cloud Storage events automatically
- The sharp library handles image resizing in Node.js functions

- Service account roles must be correctly assigned for functions to access Storage and Pub/Sub
- IAM permission changes can take a few minutes to propagate
- Real world projects combine Storage, Pub/Sub, Functions, and IAM together

Challenges & How I Solved Them

- The Eventarc service agent did not exist in the project which caused the trigger creation to fail on the first attempt. Fixed by granting the Pub/Sub publisher role to the Cloud Storage service agent using `gsutil kms serviceaccount` to get the correct service account email before redeploying.
- The Cloud Run API was not enabled when first deploying the function. Fixed by typing `y` when prompted to enable the API during deployment and waiting for it to activate.
- Accidentally selected the wrong trigger type (Pub/Sub) instead of Cloud Storage when trying to set up the function trigger in the console. Switched to using the CLI to deploy the function instead which gave more control over the configuration.
- The function showed 0 items when listing via CLI after the first deploy attempt meaning it did not deploy successfully. Fixed by granting the correct IAM roles to the Cloud Storage service agent and redeploying.
- Key lesson: Always grant the necessary IAM permissions before deploying Cloud Run Functions that use Cloud Storage triggers. The `gsutil kms serviceaccount` command gets the correct service account email for the project.

Reflection

This challenge lab brought together every skill from the Cloud Storage and IAM Foundations badge into one real-world scenario. Completing all 4 tasks independently proves practical knowledge of Cloud Storage, Pub/Sub, Cloud Run Functions, and IAM working together as a full application development environment.

Resources

GCP Cloud Run Functions Docs: <https://cloud.google.com/functions/docs>
GCP Cloud Storage Docs: <https://cloud.google.com/storage/docs>
GCP Pub/Sub Docs: <https://cloud.google.com/pubsub/docs>

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